Using a Simple Model to Simulate the "Spreading of Rumors" in a Classroom

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Abstract

In the literature, the "Spreading of Rumors" in interpersonal networks is often likened to the diffusion in physical systems. We extend this analogy, and adopt two competitive elements, namely, personal individuality and bandwagon effect, to build a simple model to simulate the rumors spread (or the competition of different opinions) in a classroom.

Our work was inspired by the famous Japanese drama "Legal High". When a person is making a decision, he/she must consider his/her personal favorites (and benefits ….) and the mainstream opinion of the community he/she is in. The competition between personal individuality and bandwagon effect cause the complicated phenomenon in "Spreading of Rumors". We setup a simple model with a control parameter representing the ratio of personal individuality and bandwagon effect. Simulation of this model shows different types of opinion's diffusion, and the possibility of phase transition in uniformity of opinion.

In this meeting, we will report the idea of our model, the simulation results in one and two dimensional networks.

Keyword: Spreading of Rumors, individuality, bandwagon effect